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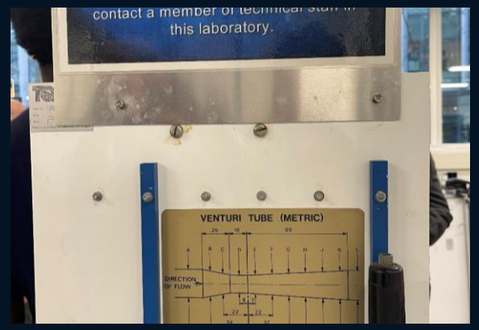
ENGINEERING PORTFOLIO / 2025 – 26

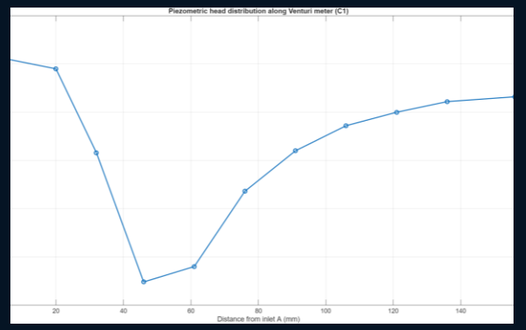
VENTURI FLOW ANALYSIS

Bernoulli's Principle I Differential-Pressure Flow Measurement

C1 BERNOULLI EXPERIMENT / C2 VENTURI & ORIFICE PLATE









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LABORATORY INVESTIGATION

Experimental study • 2026

Seeking a **2027 Industrial Placement**

INNOVATIVE ENGINEERING.
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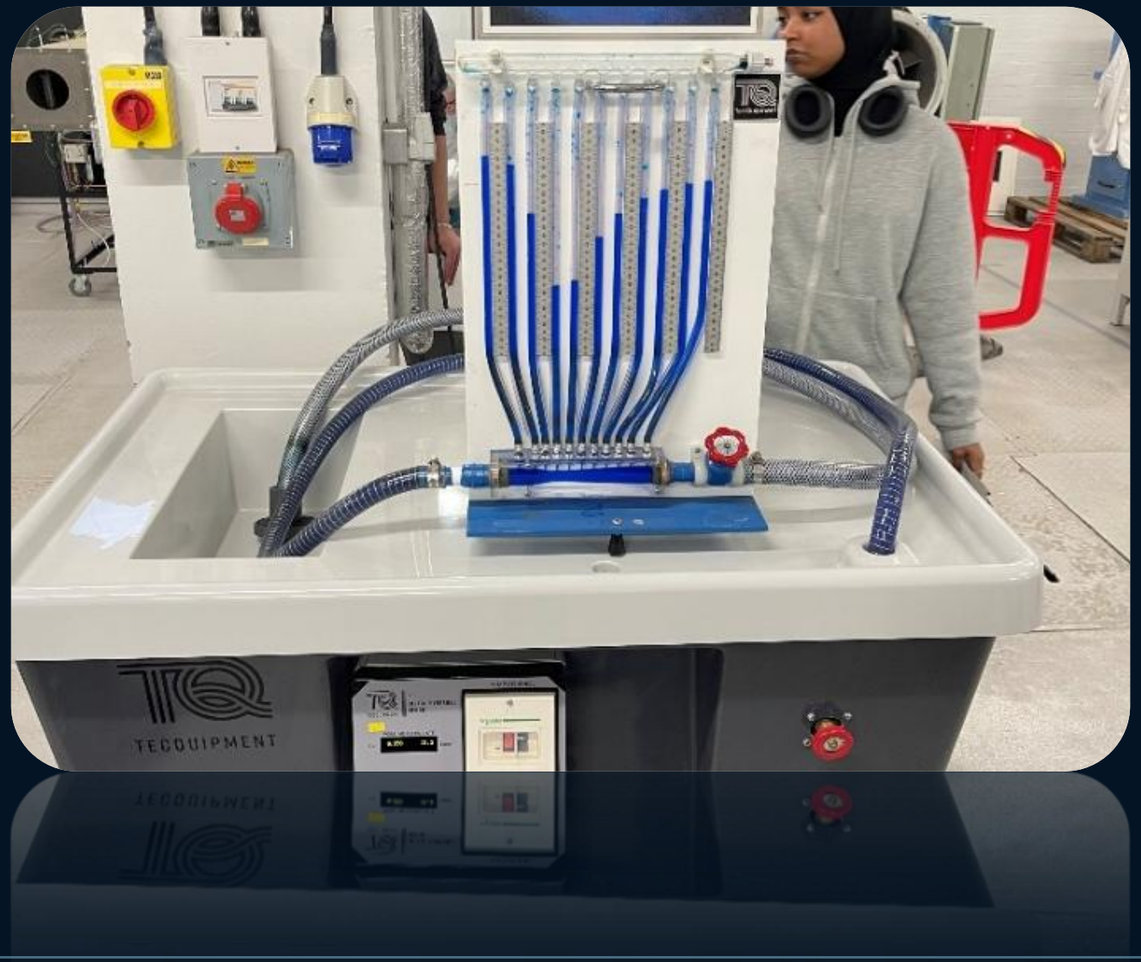
PROJECT SNAPSHOT

From theory to experimental evidence

Fluid Mechanics & Thermodynamics | Laboratory experiments C1 and C2 | 27 January 2026

The engineering challenge

Investigate how a constriction changes pressure and velocity, then compare two differential-pressure flow measurement devices using laboratory observations and calculated discharge coefficients.



PROJECT WORKFLOW

A structured experimental approach

Two linked investigations: pressure distribution first, meter performance second

01 SET UP

Inspect and level the hydraulic bench, test section and manometer.

02 MEASURE

Stabilise flow and record piezometric head and collection time.

03 CALCULATE

Apply continuity and Bernoulli-based equations to measured data.

04 INTERPRET

Plot results, compare coefficients and explain physical losses.

Inside the fluid mechanics laboratory

TecQuipment apparatus and hydraulic bench used to investigate water flow





Key apparatus

- Venturi converging–diverging test section
- Orifice plate
- Multi-tube manometer
- Gravimetric hydraulic bench
- Stopwatch and flow control valve

THEORY

Bernoulli + continuity: the physical model

Ideal-flow relationships establish a reference for real-fluid measurements

Core equations

Continuity: $Q = A \times v$

Bernoulli: $p/(\rho g) + v^2/(2g) + z = \text{constant}$

Discharge coefficient: $C_d = Q_{actual} / Q_{theoretical}$

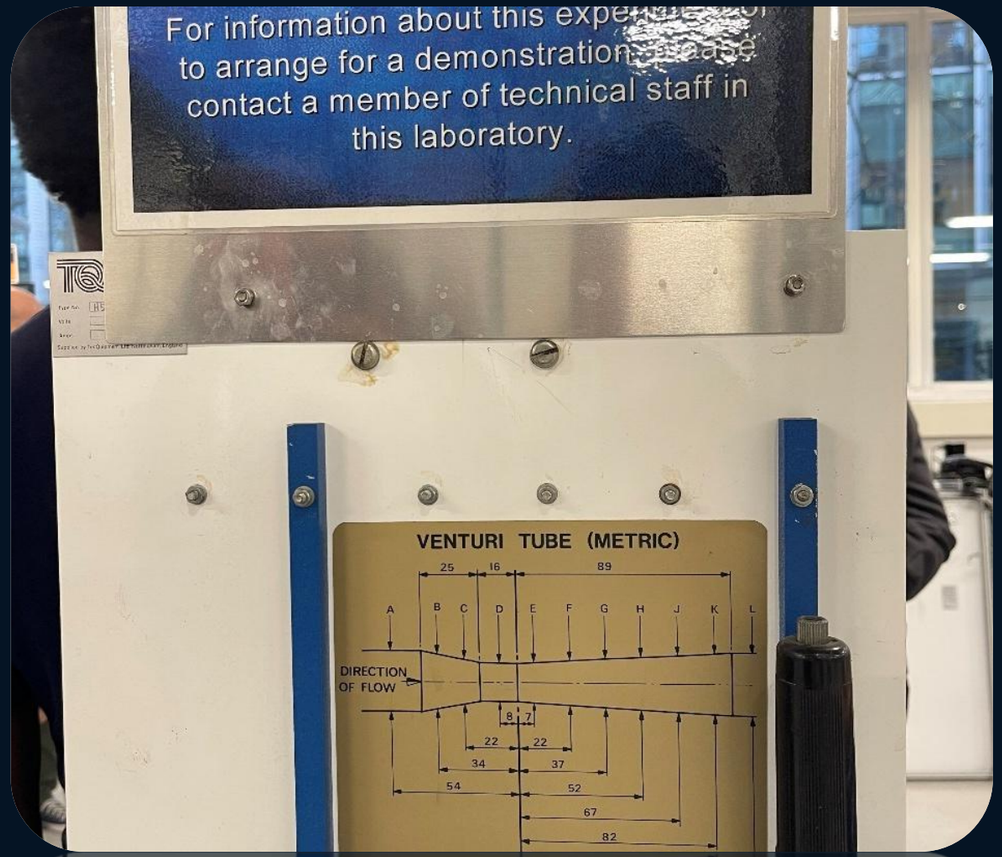
What to expect

At a constriction, cross-sectional area decreases and velocity increases. Static pressure falls at the throat. In a real fluid, friction and turbulence prevent complete downstream pressure recovery.

EXPERIMENT C1

Bernoulli’s experiment: measuring pressure head

11 tapping positions across the converging section, throat and diffuser



Experimental observations

Inlet diameter: 26 mm | Throat diameter: 16 mm

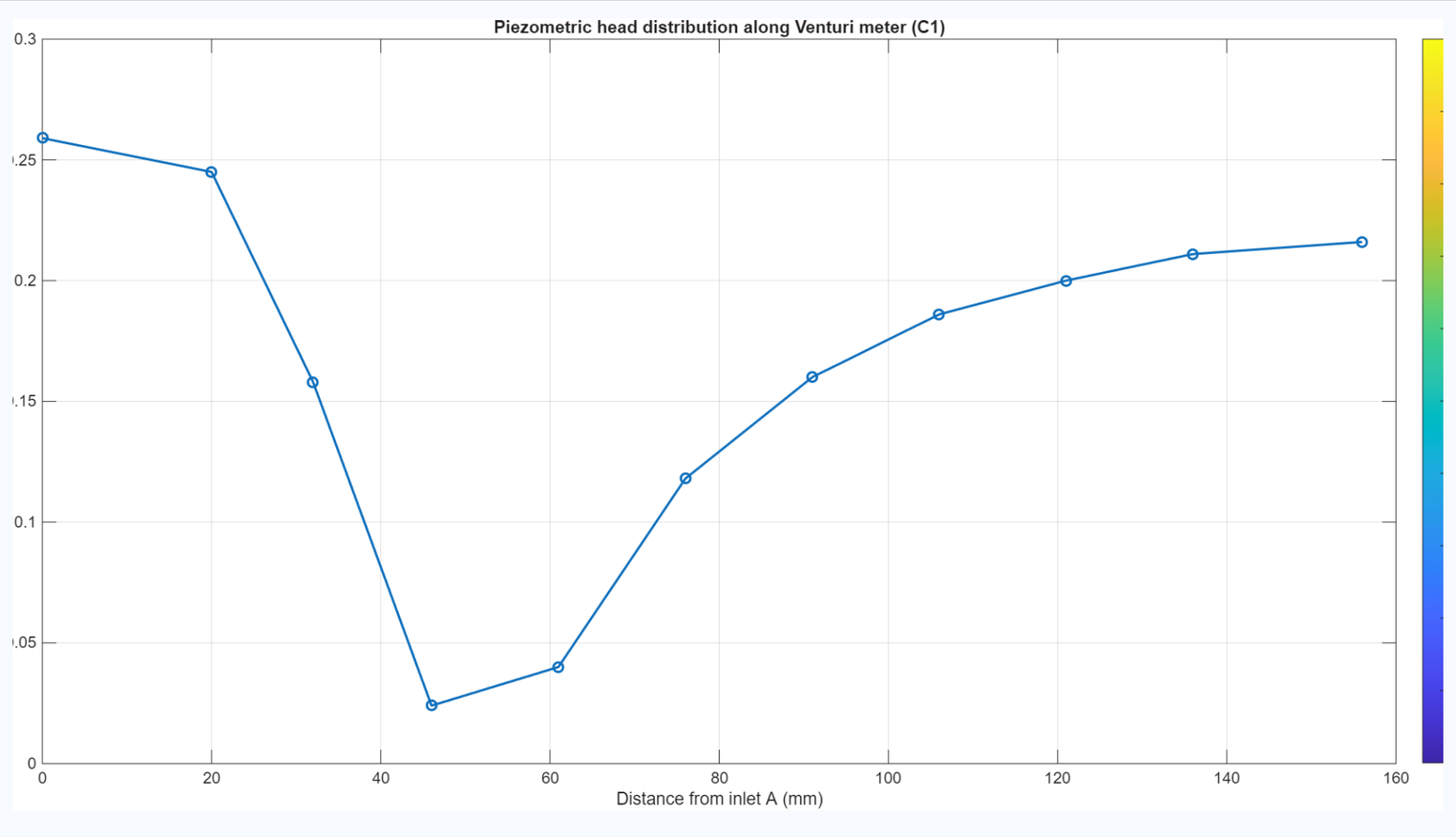
Minimum measured piezometric head: 24 mm at tapping D (46 mm from inlet).

Pressure head fell toward the throat and recovered through the diffuser, but remained below the inlet value at the final tapping.

EXPERIMENT C1

Pressure-head profile: the Venturi effect

Experimental head readings at positions A–K; plot reproduced from the submitted report



Engineering interpretation

At the throat, velocity is highest and static pressure is lowest.

The downstream head does not fully recover: inlet 259 mm; outlet 216 mm.

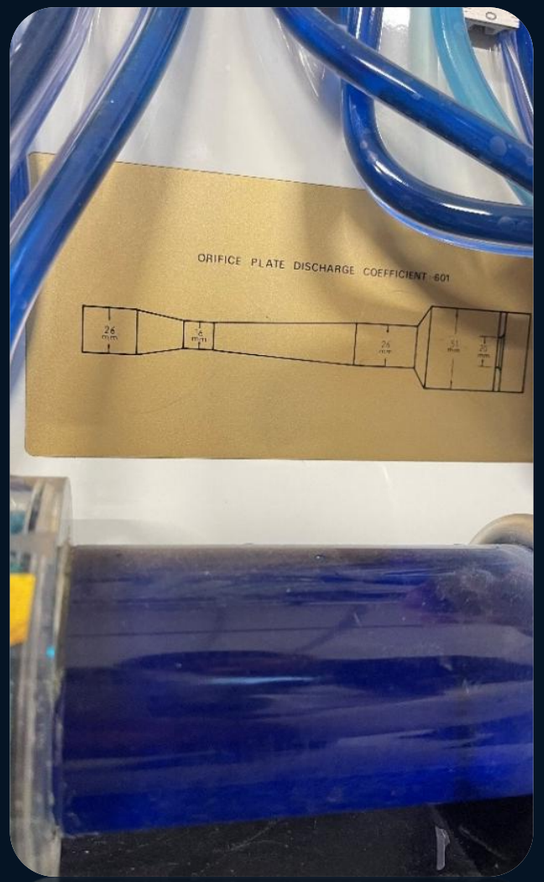
The 43 mm difference is consistent with irreversible losses in the real system.

EXPERIMENT C2

Venturi meter versus orifice plate

Comparing flow measurement characteristics at six operating points





Method

- Establish steady flow
- Record inlet and constriction head
- Collect 6 kg of water
- Time collection
- Repeat for six flow rates
- Calculate theoretical flow and Cd

EXPERIMENTAL DATA

Six flow conditions, one comparative dataset

Recorded flow rates and differential heads from the laboratory report

Test	Q (L/s)	Venturi Δh (m)	Orifice Δh (m)
1	0.490	0.276	0.315
2	0.444	0.236	0.269
3	0.401	0.187	0.215
4	0.360	0.162	0.174
5	0.318	0.112	0.132
6	0.236	0.066	0.079

Measured gravimetrically using 6 kg of water and recorded collection times.

From raw readings to discharge coefficient

Worked example — Venturi meter, test 1

Measured quantities

Mass collected = 6 kg

Collection time = 12.24 s

Actual flow rate ≈ 0.000490 m^3/s

Head difference = 361 − 85 = 276 mm = 0.276 m

Calculated result

Inlet diameter = 26 mm

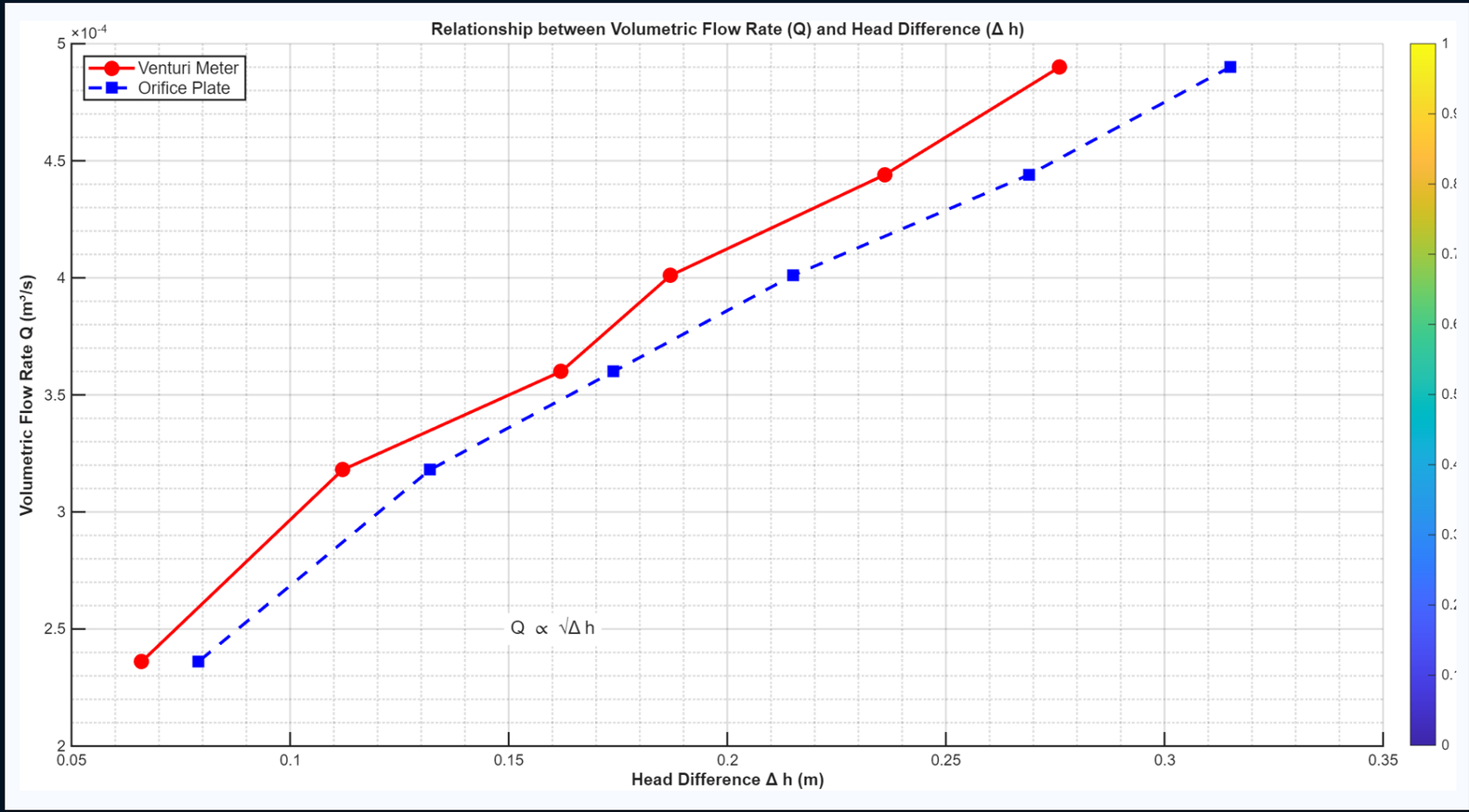
Throat diameter = 16 mm

Using Bernoulli’s equation with velocity-of-approach correction:

Venturi Cd = 0.973 (Test 1)

Flow rate versus pressure-head difference

Report graphs demonstrate the expected increasing relationship



Analysis

The report shows that volumetric flow rate rises with differential head.

Plotting the measurements helped compare the response of the two meters and connect laboratory results to the theoretical square-root relationship.

RESULTS

Quantifying meter performance

Discharge coefficients calculated for six tests in the submitted technical report

0.96	0.61	6	0.236–0.490
Venturi average Cd (approx.)	Orifice average Cd (approx.)	Flow-rate conditions	Measured Q (L/s)

Venturi meter

Higher discharge coefficient and more streamlined geometry. The measurements are closer to ideal-flow predictions.

Orifice plate

Lower discharge coefficient associated with flow contraction, turbulence and a stronger departure from ideal-flow behaviour.

CRITICAL EVALUATION

Measurement uncertainty and engineering judgement

Practical laboratory work requires more than obtaining a numerical answer

Potential sources of uncertainty

- Parallax when reading manometer levels
- Stopwatch reaction time
- Flow fluctuations before stabilisation
- Air bubbles and pressure oscillations
- Assumptions in the ideal theoretical model

What I would improve

- Repeat each test and report variability
- Use digital pressure sensors where available
- Compare measured values with uncertainty bounds
- Document flow stabilisation criteria
- Check unit conversions and calculation consistency

SKILLS DEMONSTRATED

Building the foundations of an engineer

Evidence of practical engagement, quantitative reasoning and technical communication

HANDS-ON TESTING

Setting up equipment and recording measurements.

ENGINEERING ANALYSIS

Applying fluid mechanics theory to measured results.

MATLAB VISUALISATION

Interpreting pressure-head and flow-rate graphs.

TECHNICAL REPORTING

Presenting calculations, evidence and limitations.

Applying classroom theory to real engineering

BEng (Hons) Mechanical Engineering | London South Bank University

BOKHTEAR UDDIN BADHON

Seeking a 2027 industrial placement

Interests: fluid systems • experimental testing • data analysis • sustainable engineering

PROJECT: BERNOULLI'S PRINCIPLE / VENTURI METER / ORIFICE PLATE

